Siddhant Midha

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Education

Princeton Quantum Initiative, Princeton University — Princeton, USA
Ph.D. in Quantum Science — 2024-2029 *(expected)*

- Advisors: Prof. Dmitry Abanin, Prof. Sarang Gopalakrishnan
- GPA: 4.0/4.0
- Princeton First Year Fellowship in Natural Science & Engineering — 2024-25

IIT Bombay — Mumbai, India
Bachelors in Electrical Engineering — 2020 - 2024

- Minor in Physics
- Advisors: Prof. Sai V, Prof. Bhaskaran M, Prof. Kasturi S
- GPA: 9.77/10
- *Awards*
  - Graduated with Honors, top 3% of class — 2024
  - Undergraduate Research Distinction — 2024
  - Dhananjay Fellowship — 2023
  - Chanakya Quantum Fellowship — 2023
  - Institute Academic Prize — 2023
  - Undergraduate Research Award — 2023

Research Interests

quantum error correction, quantum simulation, quantum metrology, tensor networks

Experience

Graduate Research Assistant
Princeton Quantum Initiative, Princeton University — 2024 – present

- *Tensor networks & belief propagation*
  - Turned long-standing numerical evidence for belief propagation into rigorous guarantees, and from them provably convergent, efficient tensor network contraction algorithms [5, 6]
  - Proved the performance of TNBP end-to-end as the first graph-local PEPS algorithm, showed its *algorithmic locality*. [4]
- *Quantum error correction*
  - Developed autonomous inference machines for decoding quantum LDPC codes. [2]
- *Quantum metrology*
  - Derived the limits of continuously monitored open-system metrology via mapping to matrix product states [7]
- *Quantum many-body physics*
  - Proved sharp bounds on the ground-state energy of the SYK model by mapping it to a deterministic twisted model of bosons [3].
  - Proved that the (doubly-scaled) SYK model is *Lee-Yang*, all of its Fisher zeros are purely imaginary [1].

Research Assistant
PQUEST Lab, IIT Bombay — 2023 – 2024

- *Quantum magnetometry*
  - Showed the advantage of Bayesian reconstruction for NV-center quantum diamond microscopes [11]

Research Assistant
Nanoelectronics Lab, IIT Bombay — 2022 – 2024

- *Quantum transport*

- Developed theoretical models explaining observations in transport experiments on hybrid superconducting devices [9, 12]
- Analyzed the robustness of symmetry-protected topological phases to dephasing [10]

Research Assistant
TU Munich & MPQ, Germany 2023.05 – 2023.07

- *Quantum transduction*
  - Designed computational inverse-design protocols for silicon nanophotonic structures for microwave-to-optical transduction

Research Assistant
Aalto University, Finland 2022.05 – 2022.07

- *Machine learning for quantum science*
  - Developed reinforcement learning methods to prepare quantum states [8]

Talks

1. Beyond Belief Propagation: Cluster-Corrected Tensor Network Contraction
   (a) Google Quantum AI Tensor Network Meeting (online) ⋆ 2025
   (b) Thompson group meeting, Princeton, USA ⋆ 2025
   (c) International Conference on Theory of Quantum Computation, Communication and Cryptography (TQC) 2026
2. Taming the many-body Hilbert Space: Tensor Networks and Belief Propagation
   (a) Physics Department Seminar, IIT Bombay, India ⋆ 2026
   (b) Algorithms and Complexity Seminar, Princeton, USA ⋆ 2026
   (c) ITensor Seminar, Flatiron Institute, NYC, USA ⋆ 2026
   (d) Max-Planck Quantum Theory Group Seminar ⋆(online) 2026
3. Algorithmic locality in quantum tensor networks
   (a) Interplay between neural and tensor networks, Grenoble, France 2026
   (b) IBM Quantum Seminar, Boston, MA ⋆ 2026
4. Local autonomous inference machines for quantum LDPC codes
   (a) Princeton Quantum Initiative Group Meeting, Princeton, NJ ⋆ 2026
5. Quantum metrology of continuously monitored open systens
   (a) Quantum seminar, UC San Diego, USA 2025
   (b) Quantum Engineering Seminar, IIT Bombay, India ⋆ 2026
   (c) Relativistic Quantum Information Online Circuit 2026
6. Solid-state entanglement generation with Cooper-pair splitters
   (a) APS march meeting, Minneapolis, MN, USA 2024

⋆ denotes invited talk

Publications

[1] A. Basu*, **SM***, "Lee-Yang Theorems for $q-$deformed oscillators and doubly-scaled SYK" [arXiv:2611.03624]

[2] **SM**, D. Abanin "Local autonomous inference machines for quantum LDPC codes" [arXiv:2609.38139]

[3] A. Basu*, P. Kothari*, **SM*** "Sharp bounds on the ground state energy of the SYK model" [arXiv:2607.27185]

[4] **SM**, Y. Zhang, D. Malz, D. Abanin, S. Gopalakrishnan "Algorithmic locality of Quantum Tensor Network Belief Propagation " [arXiv:2604.21919, Physical Review B]

[5] **SM***, G. Sommers*, J. Tindall, D. Abanin "Belief Propagation & Tensor Network Expansions for Many-Body Quantum Systems: Rigorous Results and Fundamental Limits" [arXiv:2604.03228]

[6] **SM**[*], Y. Zhang[*] "Beyond Belief Propagation: Cluster-Corrected Tensor Network Contraction with Exponential Convergence"
[arXiv:2510.02290, PRX Quantum 7, 033010, TQC 2026]

[7] **SM**, S. Gopalakrishnan "Metrology of open quantum systems from emitted radiation"
[arXiv:2504.13815, PRA Letter 112, L060601]

[8] K. Agaram, **SM**, V. Garg "Preparing stabilizer states via path-aware reinforcement learning" [Neurips Machine Learning for Physical Sciences, APS Global Summit 2025]

[9] A. Arora[*], **SM**[*], A. Zyuzin, P. Hakonen, B. Muralidharan,
"Steady-state dynamics and correlations in quantum-dot Cooper pair splitters"
[arXiv:2406.06053, npj Quantum Information 11, 40 (2025), APS March Meeting 2024]

[10] **SM**, K. Jana, B. Muralidharan,
"Are Symmetry Protected Topological Phases Immune to Dephasing? — A topological electronics perspective" [arXiv:2305.11149 , J. Phys. D: Appl. Phys. 57 135308]

[11] **SM**, M. Parashar, A. Bhatla, D. Broadway, J.P. Tetienne, K. Saha,
"Optimized Current Density Reconstruction from Widefield Quantum Diamond Magnetic Field Maps" [arXiv:2402.17781, Phys. Rev. Applied 22, 014015]

[12] **SM**, R. Singh, K. Gharavi, J. Baugh, B. Muralidharan,
"On the microscopics of proximity effects in one-dimensional superconducting hybrid systems" [arXiv:2411.12733]

[*]denotes equal contribution or alphabetical ordering

Awards

- **DAAD-WISE Fellowship**, German Academic Exchange Service 2023
- **MITACS Globalink Fellowship**, Canada (unavailed) 2023
- **Aalto Science Fellowship**, Aalto University 2022
- **All India Rank 150**, the JEE Advance exam (top 0.09%) 2020
- **KVPY Award**, Govt. of India (unavailed) 2020

Teaching

1. Advanced Quantum Mechanics, Princeton 2026
2. Information theory and applications, Princeton 2025
3. Quantum information and computing, IIT Bombay 2024
4. Calculus I, IIT Bombay 2022
5. Complex Analysis, IIT Bombay 2022
6. Differential equations, IIT Bombay 2021
7. Linear algebra, IIT Bombay 2021
8. Calculus II, IIT Bombay 2021, 2022

Skills

Programming: Python, Julia
Libraries: QuTiP, Qiskit, ITensor, Stim, PyTorch
Analysis: Mathematical proofs, optimization, information theory, statistical & data analysis

Mentorship

Mentored Krishna Agaram, junior undergraduate at IIT Bombay
Institute Student Mentor: to twelve freshmen at IIT Bombay
Department Academic Mentor: to eight sophomores at IIT Bombay

Service

Journal referee Physical Review Letters, Physical Review A
Conference referee for NeurIPS machine learning for physical sciences 2024, 2025
Quantum outreach

1. Mentored 12+ junior undergraduates in quantum computing research at IIT Bombay [Learning (with) quantum computers, Quantum machine learning]
2. Princeton Undergrads in Quantum Talk A conversation on quantum information

Co-organizer of the quantum information and computing group at IIT Bombay 2023-24

Co-organizer of the quantum error correction journal club at Princeton 2025-

Co-organizer of the Princeton many-body reading group 2026-